

Machine Learning Fundamentals

Week 2 – Predict Titanic Survival Using Logistic Regression

Dataset: Titanic – Machine Learning from Disaster

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Track: Neurofive Machine Learning Fundamentals Track

Introduction

After performing Exploratory Data Analysis and cleaning the Titanic dataset in Week 1, the next step is to develop a Machine Learning model capable of predicting passenger survival.

This task introduces Logistic Regression, a supervised classification algorithm used to predict binary outcomes. The objective is to preprocess the dataset, train a classification model, evaluate its performance, and interpret the prediction results using a confusion matrix.

Objectives

- Load the cleaned Titanic dataset
- Encode categorical features
- Split the dataset into training and testing sets
- Train a Logistic Regression classifier
- Predict passenger survival
- Evaluate model accuracy
- Generate and interpret a confusion matrix

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_csv(r"D:\Machine Learning Fundamentals Course\Week 1\titanic\train.csv")
```

Dataset Overview

Before training the model, the dataset is inspected to verify its structure and identify any remaining missing values.

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null   int64
1   Survived         891 non-null   int64
2   Pclass          891 non-null   int64
3   Name             891 non-null   str
4   Sex              891 non-null   str
5   Age              714 non-null   float64
6   SibSp            891 non-null   int64
7   Parch           891 non-null   int64
8   Ticket           891 non-null   str
9   Fare             891 non-null   float64
10  Cabin            204 non-null   str
11  Embarked         889 non-null   str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

Handling Missing Values

The same preprocessing approach from Week 1 is applied to ensure the dataset is complete before model training.

```
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
df["Cabin"] = df["Cabin"].fillna("Unknown")
```

Encoding Categorical Features

Machine Learning models cannot process text directly. Therefore, categorical variables are converted into numerical values using one-hot encoding.

```
df = pd.get_dummies(df, columns=["Sex", "Embarked"], drop_first=True)
```

Selecting Features and Target Variable

The target variable is **Survived**, while the remaining selected attributes are used as predictor variables.

```
X = df.drop(["Survived", "Name", "Ticket", "Cabin", "PassengerId"],
axis=1)

y = df["Survived"]
```

Splitting the Dataset

The dataset is divided into training and testing subsets using an 80:20 ratio.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

Training the Logistic Regression Model

The Logistic Regression classifier is trained using the training dataset.

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
LogisticRegression(max_iter=1000)
```

Making Predictions

The trained model is used to predict passenger survival on the testing dataset.

```
y_pred = model.predict(X_test)
```

Model Accuracy

Accuracy measures the proportion of correctly classified passengers.

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", accuracy)
```

```
Accuracy: 0.8100558659217877
```

Confusion Matrix

The confusion matrix summarizes the classification results by comparing predicted labels with actual labels.

```
cm = confusion_matrix(y_test, y_pred)
```

```
print(cm)
```

```
[[90 15]  
 [19 55]]
```

```
plt.figure(figsize=(6,5))
```

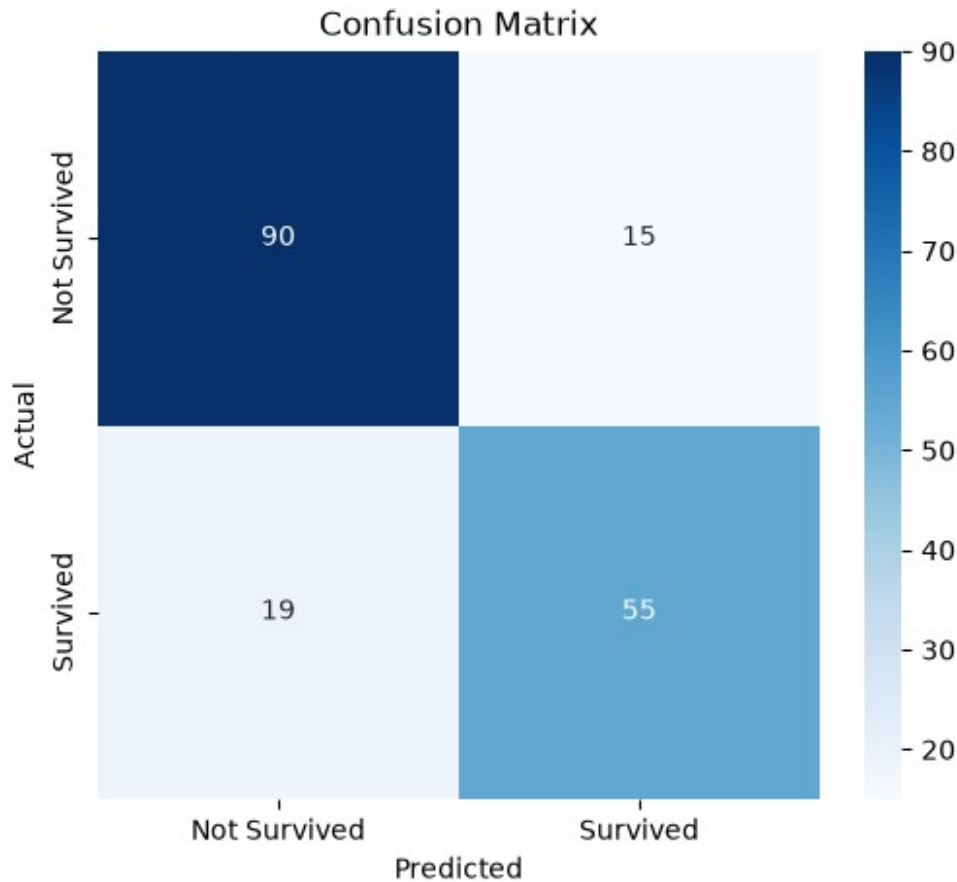
```
sns.heatmap(cm,  
            annot=True,  
            fmt='d',  
            cmap='Blues',  
            xticklabels=["Not Survived", "Survived"],  
            yticklabels=["Not Survived", "Survived"])
```

```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.title("Confusion Matrix")
```

```
plt.show()
```



Confusion Matrix Interpretation

The confusion matrix shows how many passengers were correctly and incorrectly classified.

- True Positives represent passengers who survived and were correctly predicted.
- True Negatives represent passengers who did not survive and were correctly predicted.
- False Positives indicate passengers incorrectly predicted to survive.
- False Negatives indicate passengers who survived but were predicted not to survive.

A good classification model aims to maximize true predictions while minimizing false predictions.

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.83	0.86	0.84	105
1	0.79	0.74	0.76	74

accuracy			0.81	179
macro avg	0.81	0.80	0.80	179
weighted avg	0.81	0.81	0.81	179

Conclusion

A Logistic Regression classifier was successfully trained to predict Titanic passenger survival.

Categorical variables were encoded using one-hot encoding, and the dataset was divided into training and testing subsets. The trained model achieved satisfactory classification accuracy, demonstrating that Logistic Regression provides a simple yet effective baseline model for binary classification problems.

The confusion matrix further illustrates the strengths and limitations of the model by highlighting correctly and incorrectly classified instances.